
CSE402: Numerical Analysis, Simulation and Modeling
Sessional | Section: A1 | Duration: 20 Minutes

Monte Carlo Estimation of π Using Buffon's Needle

In this assignment, you will implement a **Linear Congruential Generator (LCG)** and use it to perform a Monte Carlo simulation based on the classical **Buffon's Needle** experiment.

Your goal is to estimate π using randomly generated needle positions and orientations, and then investigate the performance of your simulation.

Task 1: 1

Use the following **Linear Congruential Generator**:

$$X_{n+1} = (aX_n + c) \bmod m$$

with the following parameters:

$$m = 10000, \quad a = 21, \quad c = 1.$$

Use the following seed:

$$X_0 = 2500$$

Implement an LCG that generates pseudo-random numbers and converts them to the range $[0, 1)$ using

$$U_n = \frac{X_n}{m}.$$

Important

Do not use Python's `random` module.

You will need **two independent random values** for each simulated needle: one for its *position* and another for its *orientation*.

Task 2: 1

Consider parallel lines separated by a distance D . A needle of length L is dropped randomly between the lines.

Use:

$$L = 1, \quad D = 2.$$

Since $L \leq D$, the probability that the needle crosses a line is:

$$P = \frac{2L}{\pi D}.$$

For each simulated needle:

1. Generate a random **position** x between 0 and $D/2$.
2. Generate a random **orientation angle** θ between 0 and $\pi/2$.
3. Determine whether the needle **crosses one of the parallel lines**.

Crossing Condition

A needle crosses a line when

$$x \leq \frac{L}{2} \sin(\theta).$$

Run the experiment for

$$N = 1,000, \quad 10,000, \quad 100,000$$

needles.

For each value of N , calculate the **estimated crossing probability**:

$$\hat{P} = \frac{\text{number of crossing needles}}{N}$$

and estimate π using:

$$\hat{\pi} = \frac{2L}{D\hat{P}}.$$

The true value of π is approximately:

$$\pi = 3.14159265.$$

Calculate the **absolute error**:

$$\text{Error} = |\hat{\pi} - \pi|.$$

Report your results:

N	Crossings	$\hat{\pi}$	Absolute Error
1,000	310	3.2258	0.08421
10,000	3199	3.1308	0.01078
100,000	31940	3.1308	0.01078

Task 3: 1

Based on your results, answer the following questions:

Reflection Questions

1. **Does increasing the number of simulations improve the estimate of π ?**
Yes, but only up to a certain number of iterations.
2. **How does the error change as N increases?**
Error decreases as iterations increase.

3. **Does the behavior of your simulation match what you would normally expect from a Monte Carlo method?**

Suppose you increase the number of simulations from 100,000 to 1,000,000. Would you expect the estimate of π to necessarily become more accurate? Explain your answer based on your investigation of the random number generator. *There might be an issue because the LCG is periodic — the sequence of pseudo-random numbers will eventually repeat. Increasing N beyond the period of the generator will not yield new randomness, and the estimate may stagnate or degrade.*